

Multivariable Calculus Final Project

Parrie, Keshav, Diego, and Michael

Documentation

Introduction & Design

For this project, our group chose a cone with radius 30" and height 36" (or 2.5' and 3', respectively). The equation that we used was

$$\frac{x^2}{900} + \frac{y^2}{900} = \frac{z^2}{1296},$$

where $x \in [-30, 30]$, $y \in [-30, 30]$, and $z \in [0, 36]$. Our cone was domain-restricted to only positive z values, as it would be physically very difficult to construct a cone that extended into the ground.

Construction Process

Our construction got off to a rough start, with our group not acquiring the proper materials until

Calculations

Theoretical Precise Value

To calculate the precise theoretic value, we will use the triple integral

$$\iiint_Q f(x, y, z) dV.$$

To find the bounds, we will use cylindrical coordinates, as Cartesian coordinates tend to end up messy in such calculations. To find the bounds in cylindrical coordinates, we only need to solve for r , as $\theta \in [0, 2\pi]$ and $z \in [0, 36]$, as given by our domain restrictions above.

To find the bounds for r , we first convert the rectangular x and y to cylindrical coordinates. Following the conversion, our cone can be represented by the equation $\frac{r^2}{900} = \frac{z^2}{1296}$, which we can solve to find the upper bound for r :

$$\frac{r^2}{900} = \frac{z^2}{1296} \implies r^2 = \frac{900z^2}{1296} \implies r = \sqrt{\frac{900z^2}{1296}} = \sqrt{\frac{25z^2}{36}} = \frac{5z}{6}.$$

Thus, the upper bound for r is $\frac{5z}{6}$, and because the lower bound for r is 0, we know that the complete bounds for r are $[0, \frac{5z}{6}]$. Thus, the complete bounds, for all variables, are:

$$\theta \in [0, 2\pi], \quad z \in [0, 36], \quad \text{and} \quad r \in \left[0, \frac{5z}{6}\right].$$

Therefore, the volume integral for the cone is If we plug in all the values for the triple integral, we end up with

$$\int_0^{2\pi} \int_0^{36} \int_0^{\frac{5z}{6}} r \, dr \, dz \, d\theta,$$

which we can solve as follows:

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^{36} \int_0^{\frac{5z}{6}} r \, dr \, dz \, d\theta = \int_0^{2\pi} \int_0^{36} \left[\frac{r^2}{2} \right]_0^{\frac{5z}{6}} dz \, d\theta = \int_0^{2\pi} \int_0^{36} \frac{25z^2}{72} dz \, d\theta \\ &= \int_0^{2\pi} \left[\frac{25z^3}{216} \right]_0^{36} d\theta = 5040 \int_0^{2\pi} d\theta = 5040 (2\pi) = 10800\pi \approx 33929.2007. \end{aligned}$$

Thus, the theoretical precise volume of our cone is 33929.2007 cubic inches (or approximately 19.6348 cubic feet).